

$$R = R_y(\psi) \cdot R_z(\phi) \cdot R_z(\theta)$$

$$= \begin{matrix} \xleftarrow{R_y(\psi)} & R_x(\phi) & R_z(\theta) \end{matrix} \begin{bmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{bmatrix} \begin{bmatrix} 1 & \cos\theta & -\sin\theta & 0 \\ \cos\phi \sin\theta & \cos\phi \sin\theta & -\sin\phi \\ \sin\phi \sin\theta & \sin\phi \cos\theta & \cos\phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos\psi \cos\theta + \sin\psi \sin\phi \sin\theta & -\cos\psi \sin\theta + \sin\psi \sin\phi \cos\theta & \sin\psi \cos\phi \\ \cos\phi \sin\theta & \cos\phi \sin\theta & -\sin\phi \\ -\sin\psi \cos\theta + \cos\psi \sin\phi \sin\theta & \sin\psi \sin\theta + \cos\psi \sin\phi \cos\theta & \cos\psi \cos\phi \end{bmatrix}$$

$\phi = \pi/2$ 라면. $\cos\phi = 0$, $\sin\phi = 1$

$$R = \begin{bmatrix} \cos\psi \cos\theta + \sin\psi \sin\theta & \sin\psi \cos\theta & 0 \\ 0 & 0 & -1 \\ -\sin\psi \cos\theta + \cos\psi \sin\theta & \sin\psi \sin\theta + \cos\psi \cos\theta & \cos\psi \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\psi - \theta) & \sin(\psi - \theta) & 0 \\ 0 & 0 & -1 \\ \sin(\psi - \theta) & \cos(\psi - \theta) & 0 \end{bmatrix}$$

→ $(\psi - \theta)$ 로 회전 운동을 서술하게 되며,

ψ 를 변경하는 것과 θ 를 변경하는 것 사이에

차이가 없게 되었다. = 겹쳐짐